Paper 2 · Answers & Solutions

2026 · 45 questions

Part 1 · Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	D	D	A	B	A	A	C	C	D	A	B	C	A	D
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A	C	C	D	A	B	B	C	A	C	C	A	D	A	D
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
B	D	B	D	B	C	B	D	B	C	B	B	D	A	C

Part 2 · Worked Solutions

Section A

1 **A** — Algebraic Operations

1. $(2x-y)^2 = 4x^2 - 4xy + y^2$
2. $(x+2y)(x-2y) = x^2 - 4y^2$
3. The expression $= 3x^2 - 4xy + 5y^2$; the answer is A

2 **D** — Laws of Indices

1. $(3a^{-2}b)^2 = 9a^{-4}b^2$
2. $\dfrac{9a^{-4}b^2}{9a^3b^{-4}} = a^{-7}b^6$
3. $= \dfrac{b^6}{a^7}$; the answer is D

3 **D** — Change of Subject

1. $a - 2b = 3bc$
2. $a = 3bc + 2b = b(3c+2)$
3. $b = \dfrac{a}{3c+2}$; the answer is D

4 **A** — Approximation

1. Correct to 2 sig. fig.: $\mathbf{0.031}$; 3 sig. fig.: $\mathbf{0.0307}$
2. Correct to 2 d.p.: $\mathbf{0.03}$; 4 sig. fig.: $\mathbf{0.03065}$
3. Only '$\mathbf{0.031}$ (2 sig. fig.)' is true; the answer is A

5 **B** — Simultaneous Equations

1. Let each pen cost $\$p$ and each notebook $\$n$: $3p + 2n = 46$, $2p + 5n = 82$
2. Solving, $p = 6$, $n = 14$
3. $4(6) + 3(14) = \$66$; the answer is B

6 **A** — Identities

1. L.H.S. $= (a+b)x^2 + (2a-2b)x + a + b + c$
2. $a + b = 5$, $a - b = 1 \Rightarrow a = 3$, $b = 2$; $c = 6 - 5 = 1$
3. $a : b : c = 3 : 2 : 1$; the answer is A

7 **A** — Quadratic Equations

1. Let $u = x - k$: $u^2 - 3u - 10 = 0$
2. $(u-5)(u+2) = 0$, $u = 5$ or -2
3. $x = k + 5$ or $k - 2$; the answer is A

8 **C** — Functions

1. $1 + c + 5 = 4 + 2c + 5$
2. $c = -3$
3. $g(4) = 16 - 12 + 5 = 9$; the answer is C

9 **C** — Remainder Theorem

1. $f(-1) = -3 - k - 1 + 2 = 0 \Rightarrow k = -2$
2. Remainder $= f(2) = 24 + 8 + 2 + 2$
3. $= 36$; the answer is C

10 **D** — Inequalities

1. $2x - 3 > 5 \Rightarrow x > 4$
2. $\dfrac{x}{3} \leq -1 \Rightarrow x \leq -3$
3. The solution is $x \leq -3$ or $x > 4$; the answer is D

11 **A** — Percentages

1. Non-local boys $= 60\% \times 15\% = 9\%$, non-local girls $= 40\% \times 20\% = 8\%$
2. Required $= \dfrac{8}{9+8}$
3. $\approx 47.1\%$; the answer is A

12 **B** — Rates

1. Time taken $= \dfrac{120}{60} + \dfrac{120}{40} = 5$ hours
2. Total distance $= 240$ km
3. Average speed $= \dfrac{240}{5} = 48$ km/h; the answer is B

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13 C — Variations

1. $z = \dfrac{kx^2}{y}$
2. New $z = \dfrac{k(1.2x)^2}{0.9y} = 1.6z$
3. z is increased by 60% ; the answer is C

14 A — Graphs of Quadratic Functions

1. It opens downwards and the vertex is $(-1, 4)$
2. y-intercept $= -1 + 4 = 3$
3. When $x = 1$, $y = -4 + 4 = 0$, so it passes through $(1, 0)$; the answer is A

15 D — Arc Lengths and Areas of Sectors

1. Area $= \dfrac{1}{2}r\times$ arc length: $27\pi = \dfrac{1}{2}r(6\pi) \Rightarrow r = 9$ cm
2. $\dfrac{\theta}{360^\circ} \times 2\pi(9) = 6\pi$
3. $\theta = 120^\circ$; the answer is D

16 A — Volumes of Solids

1. $\dfrac{\pi r^2 h}{\frac{1}{3}\pi(4^2)(2h)} = \dfrac{3}{8}$
2. $\dfrac{3r^2}{32} = \dfrac{3}{8} \Rightarrow r^2 = 4$
3. $r = 2$; the answer is A

17 C — Similar Triangles and Areas

1. $\dfrac{AE}{AB} = \dfrac{AF}{AD} = \dfrac{1}{3}$, $\triangle AEF \sim \triangle ABD$, $EF \parallel BD$
2. Area of $\triangle ABD = 2 \times 3^2 = 18$ cm^2
3. Area of trapezium $EBDF = 18 - 2 = 16$ cm^2 ; the answer is C

18 C — Pythagoras' Theorem

1. $AC = \sqrt{3^2 + 4^2} = 5$ cm
2. $AD = \sqrt{5^2 + 12^2} = 13$ cm
3. Perimeter $= 3 + 4 + 12 + 13 = 32$ cm; the answer is C

19 D — Parallel Lines and Angles

1. Draw lines through C and D parallel to AB
2. $\angle BCD = 35^\circ + \beta \Rightarrow \beta = 45^\circ$; $\angle CDE = \beta + (180^\circ - 150^\circ)$
3. $\angle CDE = 45^\circ + 30^\circ = 75^\circ$; the answer is D

20 A — Polygons

1. $(n - 2)180^\circ = 1440^\circ \Rightarrow n = 10$
2. Number of diagonals $= \dfrac{10(10 - 3)}{2} = 35$; I is true
3. A regular decagon has 10 axes of reflectional symmetry; II is true; its order of rotational symmetry is 10 ; III is false
4. Only I and II are true; the answer is A

21 B — Properties of Quadrilaterals

1. The diagonals of a rhombus bisect the interior angles, so $\angle ABD = \angle CBD$; I is true
2. AC is the perpendicular bisector of BD and also of FE (rectangle), so C is equidistant from F and E ; III is true
3. In general $AF \neq CE$ (e.g. $AF = \sqrt{109}$, $CE = \sqrt{13}$); II is not necessarily true
4. Only I and III must be true; the answer is B

22 B — Properties of Circles

1. In $\triangle AOF$, $\angle OAF = 25^\circ$ and $\angle AOF = 70^\circ$
2. $\angle AFC$ is an exterior angle of $\triangle AOF$
3. $\angle AFC = 25^\circ + 70^\circ = 95^\circ$; the answer is B

23 C — Trigonometric Ratios

1. $AB = AC\cos\alpha$
2. $AC = AD\cos\beta$
3. $AB = AD\cos\alpha\cos\beta$, so $AB : AD = \cos\alpha\cos\beta : 1$; the answer is C

24 A — Transformations in Coordinates

1. Anticlockwise rotation through $90°$: $(x, y) \to (-y, x)$, giving $(2, 3)$
2. Reflection in $x = 5$: $x \to 10 - x$
3. The image is $(8, 3)$, with x-coordinate 8; the answer is A

25 C — Coordinate Geometry

1. Perpendicular bisector of AB: mid-point $(3, 2)$, slope -1, $y = -x + 5$
2. $x + 1 = -x + 5 \Rightarrow x = 2$
3. $y = 3$; the answer is C

26 C — Equations of Straight Lines

1. Not intersecting means parallel: $\frac{k-1}{3} = \frac{2}{k}$
2. $k^2 - k - 6 = 0 \Rightarrow k = 3$ or -2 (neither gives coincident lines)
3. $k = -2$ or 3; the answer is C

27 A — Equations of Circles

1. Dividing by 2: $x^2 + y^2 - 4x + 6y - 3 = 0$, centre $(2, -3)$, radius $= \sqrt{4 + 9 + 3} = 4$
2. Substituting the origin gives $-3 < 0$, so the origin lies inside; I is false
3. Circumference $= 8\pi$; II is true. Distance from centre to y-axis $= 2$; III is true
4. Only II and III are true; the answer is A

28 D — Probability

1. There are $C_2^6 = 15$ equally likely outcomes
2. Product not less than 12: $\{2, 6\}, \{3, 4\}, \{3, 5\}, \{3, 6\}, \{4, 5\}, \{4, 6\}, \{5, 6\}$
3. Probability $= \frac{7}{15}$; the answer is D

29 A — Measures of Dispersion

1. From the diagram, least value $= 10$, $Q_1 = 18$, $Q_3 = 30$, inter-quartile range $= 12$
2. $m - 10 = 3(12)$
3. $m = 46$; the answer is A

30 D — Central Tendency and Dispersion

1. $a + b = 14$, so one of a, b is at most 7 and the other at least 7; the median must be 7; I is true
2. If $a = 4$, $b = 10$, the modes are 4 and 10; II is not necessarily true
3. If $a = 1$, $b = 13$, the standard deviation $= \sqrt{18} \approx 4.24 > 3$; III is not necessarily true
4. Only I must be true; the answer is D

Section B

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31 B — H.C.F.

1. H.C.F. of the numerical parts $= 6$
2. Take the lowest powers: x^2 , y^2 ; $18x^2y^4$ has no z
3. H.C.F. $= 6x^2y^2$; the answer is B

32 D — Number Systems

1. $\mathrm{B} = 11$, $\mathrm{C} = 12$
2. $5\mathrm{B0C}_{16} = 5 \times 16^3 + 11 \times 16^2 + 0 \times 16 + 12$
3. $= 5 \times 16^3 + 11 \times 16^2 + 12$; the answer is D

33 B — Logarithms

1. $\log(xy) = 1 \Rightarrow xy = 10$
2. $(y+3)y = 10 \Rightarrow (y-2)(y+5) = 0$
3. $y > 0$, so $y = 2$ ($y = -5$ rejected); the answer is B

34 D — Graphs of Logarithmic Functions

1. Slope $= \dfrac{-1-3}{2-0} = -2$, $y = 3 - 2\log x$
2. $\log 1000 = 3$
3. $y = 3 - 6 = -3$; the answer is D

35 B — Complex Numbers

1. $(k+2i)(3-i) = 3k - ki + 6i - 2i^2$
2. $= (3k+2) + (6-k)i$
3. Imaginary part 0 : $6 - k = 0$, $k = 6$; the answer is B

36 C — Sequences

1. $T(1) = S(1) = 5$; I is true
2. $T(n) = S(n) - S(n-1) = 2(2n-1) + 3 = 4n +$ (also true for $n = 1$); II is true
3. $T(n+1) - T(n) = 4$, so it is arithmetic; III is true
4. I, II and III are all true; the answer is C

37 B — Linear Programming

1. The vertices of R are $(0,1)$, $(6,1)$ and $(0,7)$
2. The values of $2x - y + c$ are $c-1$, $c+11$, $c-7$; the least is $c-7$
3. $c - 7 = -4 \Rightarrow c = 3$; the answer is B

38 D — Properties of Circles (Tangents)

1. $\angle TAB = \angle ACB = 40^\circ$ ($\angle$ in alt. segment)
2. In $\triangle TAC$, $\angle TAC = 180^\circ - 30^\circ - 40^\circ = 110^\circ$
3. $\angle BAC = 110^\circ - 40^\circ = 70^\circ$; the answer is D

39 B — Trigonometric Equations

1. $(2\sin\theta + 1)(\sin\theta - 1) = 0$
2. $\sin\theta = 1 \Rightarrow \theta = 90^\circ$; $\sin\theta = -\dfrac{1}{2} \Rightarrow \theta = 210^\circ$ or 330°
3. 3 roots in total; the answer is B

40 C — Trigonometry in 3-D

1. $BG = \dfrac{2}{3} \times 6\sin 60^\circ = 2\sqrt{3}$ cm,
   $AB = \sqrt{(2\sqrt{3})^2 + 4^2} = \sqrt{28}$ cm
2. $AB = AC$, so the foot of perpendicular from A to BC is the mid-point of BC
3. $\cos\angle ABC = \dfrac{3}{\sqrt{28}} = \dfrac{3\sqrt{7}}{14}$; the answer is C

41 **B** — Centres of Triangles

1. $\triangle ABC$ is obtuse-angled, so its orthocentre lies outside the triangle; I is false
2. The circumcentre of an obtuse-angled triangle lies outside the triangle; II is true
3. The centroid and in-centre coincide only in an equilateral triangle; III is false
4. Only II is true; the answer is B

42 **B** — Permutation and Combination

1. Arrange the boys first: $4! = 24$
2. Arrange the girls in the 5 gaps: $P_3^5 = 60$
3. $24 \times 60 = 1440$; the answer is B

43 **D** — Probability

1. Probability that all fail $= 0.5 \times 0.4 \times 0.2 = 0.04$
2. At least one solves $= 1 - 0.04$
3. $= 0.96$; the answer is D

44 **A** — Measures of Dispersion

1. Median $= 8$; I is false
2. Mean $= 8$, standard deviation $= \sqrt{\dfrac{25+9+0+4+36}{5}} = \sqrt{14.8} \approx 3.85$; III is true
3. Standard score of $14 = \dfrac{14-8}{3.85} \approx 1.56 < 2$; II is false
4. Only III is true; the answer is A

45 **C** — Measures of Dispersion

1. Multiplying each datum by -2 multiplies the standard deviation by $|-2| = 2$
2. Adding 3 does not change the standard deviation
3. New standard deviation $= 2 \times 4 = 8$; the answer is C